

Project Proposal – Milestone 1

Airline Reservation & Flight Management System

Course Name:	Data Structures & Algorithms
Submitted To::	Athar Ikhlq
Section	F
Project Title :	Airline Reservation & Flight Management System

Group Members Details

Name	Roll Number
Hafiz Dawood Shafiq	241980075
Syed Umair	241980084
Abubakar Sohail	241980004
Tehami Naveed	241980101
Muhammad Farooq Adnan	241980034

● Overview

This project focuses on developing a structured and efficient Airline Reservation & Flight Management System. It supports real-time flight scheduling, seat availability, passenger management, booking flows, and route organization. The system will simulate critical airline operations to demonstrate how core Data Structures & Algorithms provide efficient data storage, fast searching, and scalable processing in a production-like system.

◆ Introduction

The Airline Reservation & Flight Management System is designed to provide a robust back-end for managing flights and reservations while exposing simple web interfaces for passengers and administrative users. The system will include modules for searching flights, making and canceling bookings, managing passenger profiles, and mapping flight routes. Implementation will prioritize clean data structure usage (arrays, queues, stacks, hashing, and trees/graphs), modularity, and persistent storage via SQL or file-based mechanisms. The goal is to create a maintainable system that demonstrates algorithmic thinking and practical application of DSA concepts taught in the course.

✧ **Complete Project Scope**

- Manage and update flight schedules with availability and seat maps.
- Allow customers (passengers) to search, reserve, and cancel bookings.
- Provide user accounts and passenger profile management using hashing for fast lookup.
- Use queue-based flow for handling booking requests and waitlists.
- Implement stack-based cancellation history and undo operations for admins.
- Represent flight routes using trees/graphs for route planning and connection searches.
- Generate reservation and flight-related reports for administrators.
- Integrate the system with SQL/File-based storage for persistence and reporting.

❖ **List of Features (Who performs which task)**

1. Passenger:

- Create and manage personal profile
- Search flights by date, destination, and price
- Book tickets and select seats
- View booking history and boarding passes
- Request cancellations and view refund status

2. Booking Agent / Kiosk:

- Assist passengers with bookings and special requests
- Place passengers on waitlists (queue) for full flights
- Process group bookings and special fares

3. Flight Crew (Pilot / Air-hostess):

- Access assigned flight manifests
- View passenger special needs and seating arrangements
- Update flight status (on-time, delayed) for live tracking

4. Administrator:

- Create, update, and cancel flight schedules
- Manage seat inventory and fare classes
- Approve cancellations and perform undo operations
- Generate operational and financial reports
- Manage user accounts and system settings

5. System:

- Automatically assign seats based on availability
- Handle priority queueing for urgent/reserved bookings
- Map optimal routes using graph/tree modules

➤ Tentative Data Structures to be Used

Arrays - Flight schedules, seat matrices, and timetable storage.

Queues - Booking request flows, wait-lists, and processing order.

Stacks - Cancellation history and undo operations for admin actions.

Hashing - Passenger and booking lookup for $O(1)$ average access.

Trees / Graphs - Route mapping, connection searches and route hierarchies.

■ Milestone 1: 4-Week Timeline (Week 1–4)

Week 1:

- Group formation and role assignment.
- Initial requirement gathering and scope finalization.
- Stakeholder (teacher) consultation and project idea approval.

Week 2:

- Detailed system design: UML diagrams (class, sequence, use-case).
- Define data models and initial database schema.
- List detailed features and map DSAs to each module.

Week 3:

- Create sample datasets and run basic functional tests.
- Prepare first draft of proposal document with screenshots/diagrams.

Week 4:

- Refine proposal based on feedback; finalize UML diagrams and timeline.
- Prepare final document (Milestone 1) with filled group details and polished introduction.
- Practice short presentation; submit proposal and collect approval.

✓ Approved by all group members

✓ Approval signature _____ . thank you.